

Shivang Dubey

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SUMMARY

Data Engineer at ZS Associates with **2.5+** years of experience delivering ETL pipelines, analytics platforms, and backend systems for a Fortune 500 pharmaceutical client. Engineered 500+ data pipelines processing **5M+** healthcare records at **99%** accuracy, automated workflows saving 24+ hours weekly, and led backend development for a 1,000+ user platform. Python, SQL, AWS, Airflow. Currently pursuing a PG Diploma in Data Science from IIIT Delhi. Mitacs research alumnus, Polytechnique Montréal.

SKILLS SUMMARY

- **Languages:** Python, SQL (MySQL, PostgreSQL), C/C++
- **Tools & Frameworks:** pgAdmin, Numpy, Pandas, OpenCV, Flask, Git, GitHub
- **BI & Reporting Tools:** Power BI, MicroStrategy, Excel, Powerpoint
- **ETL & Orchestration:** Apache Airflow, IICS
- **Cloud & Data Platforms:** AWS (S3, EC2, ECS, Redshift), Snowflake
- **Key Skills:** Data Analysis, Data Visualization, ETL Development, Data Ingestion, Backend Development, Machine Learning, Healthcare Data (Pharma Drug Data, Claims, Physician & Market Sales Analytics), Claude, ChatGPT, & Gemini

EXPERIENCE

- **ZS Associates** Gurugram, India
Business Technology Solutions Associate Consultant Sep 2023 - Present
 - Played a key role in an Operations and Enhancement Project for the **US's** top biomedical and pharmaceutical company, driving the optimization of data processes and systems. Designed, built, and enhanced multiple **ETL** pipelines critical to the project's success.
 - Developed and executed scalable ETL processes that converted **500+** raw datasets into actionable sales and compensation reports, implementing rigorous data validation measures to achieve **99%** data accuracy across systems.
 - Conducted routine data manipulation and insight generation using SQL (PostgreSQL, Workbench). Managed data ingestion and extraction via Amazon S3, and processed large healthcare datasets including medical sales and claims data exceeding **5M+** records.
 - Developed performance analytics dashboards for a healthcare field force of over **1,000** representatives, enabling actionable insights for individual performance tracking and contributing to a **25%** increase in goal attainment.
 - Automated backup processes on Amazon S3 using Python and **boto**, reducing manual effort by approximately **8 hours** per week. Streamlined operations and reporting workflows using Apache Airflow, saving an additional 4 hours weekly.
 - Developed Excel automation solutions using VBA, eliminating a **12-hour** weekly manual process for data fetching, formatting, and analysis. Created VBA-driven Excel reports linked to PostgreSQL databases, enhancing team efficiency by approximately **80%**.
 - Led backend engineering for a web portal as part of a critical technology migration project, implementing complex business logic with Python and Flask and integrating with an AngularJS frontend. The platform supports **1000+** users in a US healthcare marketing ecosystem managing data for **4** key pharmaceutical products.
- **Polytechnique Montréal** Montréal, Canada
Summer Research Intern May 2022 - Jul 2022
 - Collaborated with a PhD student to develop a **Deep Q Learning** model for a network intrusion detection system. The obtained accuracy was greater than **79%**.
 - Built machine learning models for the same **Network Intrusion Detection System**, including **Support Vector Machine**, **Random Forest Algorithm**, and **Convolution Neural Network**, and compared the results to the Deep Q Learning model using **Tensorflow**.
 - Compiled the analysis and found that **Deep Q Learning performed better** than other ML models on the same dataset, with an accuracy difference of **2%**.

EDUCATION

- **Indraprastha Institute of Information Technology Delhi** IIIT Delhi
Post Graduate Diploma in Data Science in Health and Climate Change for Social Impact Oct 2025 - Present
- **Guru Gobind Singh Indraprastha University** BPIT Delhi
B.Tech. in Electronics and Communication Engineering; CGPA: 9.45/10.00 Aug 2019 - Jul 2023
Courses: Analog Electronics, Data Structures and Algorithms, Digital Signal Processing, Digital Communication, Database Management System

ACADEMIC PROJECTS

- **Social Distancing Violation Detection System:** The model was created to detect human and distance violations between two human centroids along with human face detection.
 - **Tech:** Model trained using **YOLO** (v3) algorithm and weights, along with COCO dataset for pre-existing object classes. **OpenCV** for face and human detection. Along with **SciPy's** Euclidean Distance for human centroid formation.
 - **Outcome:** With a suitable detection rate, the model draws a box around the human; Accuracy more than **94%**. [Thesis](#)